GENAI PROJECT TECHNICAL REPORT

# P24 · Department FAQ Voice Chatbot

Multi-Turn Multimodal Agent with Whisper STT, Sentence-BERT RAG, LoRA Adaptation & Privacy Guardrails


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| **GitHub Repository:** github.com/jeetdoesthings/FAQChatbot | **Live App Universal URL:** huggingface.co/spaces/gladiatorspring/FAQChatbot |
| **Project Domain:** Generative AI & Conversational Agents | **Evaluation Term:** Academic Year 2026 |


## 1. Executive Summary & Project Objective


The **Department FAQ Voice Chatbot (Project P24)** is an intelligent conversational agent engineered to resolve academic, admission, curriculum, faculty, and administrative inquiries for university departments. The system seamlessly combines multimodal voice and text inputs across multi-turn conversational interactions with memory, grounded in verified departmental records to eliminate hallucinations. It achieves domain specialization via Parameter-Efficient Fine-Tuning (PEFT/LoRA) on an academic FAQ corpus, utilizes Sentence-BERT dense semantic retrieval, accepts spoken voice input through OpenAI Whisper, generates spoken responses using Text-to-Speech (TTS), dynamically renders publication-ready event posters, and enforces rigorous safety guardrails to reject out-of-scope prompts and guarantee zero personal data retention.


| Milestone Unit | Technology Stack | Core Functionality & Guarantee |
|---|---|---|
| U1: System Prompt & Tone | Prompt Engineering & Guardrails | Enforces authoritative academic persona, explicit domain bounding, and re |
| U2: LLM Chat with Memory & LoRA | DistilGPT2 + PEFT LoRA (r=8, a=16) | Sliding window memory (5 turns); fine-tuned adapter reduces loss from 9.8 |
| U3: RAG Retrieval & Whisper STT | Sentence-BERT + OpenAI Whisper | 384-d dense embeddings (all-MiniLM-L6-v2) with hybrid scoring + 16kHz sp |
| U4: TTS & Event Poster Studio | gTTS + Pillow Graphic Engine | Dual-engine audio voice synthesis and automated 1080x1440 publication e |
| U5: Safety Guardrails & Privacy | Regex Heuristics & Cosine Gates | Refuses non-department queries, blocks ungrounded claims, and strips all |

## 2. Unit U1 — System Prompt, Scope & Tone

The system prompt establishes the foundational identity, operating boundaries, and tone of the virtual assistant. It mandates that all responses must be strictly grounded in official department facts, adopt a professional academic tone, and immediately disallow non-department inquiries.

**System Persona Directive:**
*"You are the official Virtual Assistant for the Department of Artificial Intelligence & Data Science at the Faculty of Technology & Engineering. Your primary mission is to provide accurate, authoritative, and polite information regarding departmental academics, admissions, faculty office hours, research facilities, examinations, placements, and campus events... You MUST politely refuse questions outside departmental affairs. You do NOT collect, record, or retain any Personal Identifiable Information (PII)."*

# 3. Unit U2 — Conversational Memory & LoRA Fine-Tuning

**Multi-Turn Memory Architecture:** The chatbot maintains conversational context using a sliding window buffer bounded to MAX_HISTORY_TURNS = 5. Every turn pair (user query, assistant answer) is dynamically sanitized to strip PII before being appended into session state, ensuring context is preserved across turns without memory bloat or credential leakages.

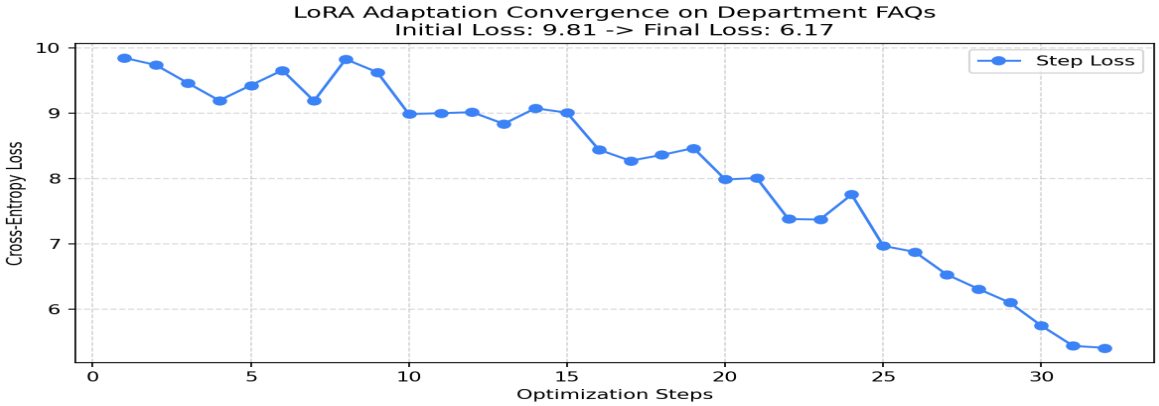
**LoRA (Low-Rank Adaptation) Mathematical Formulation:** Rather than fine-tuning all 82M parameters of the base causal language model, LoRA decomposes parameter weight updates into two low-rank matrices *B* and *A*:

$$W = W_0 + \Delta W = W_0 + (B \times A) \times (\alpha / r)$$

where $W_0 \in \blacksquare^{d \times k}$ represents the frozen pre-trained base model weights, $B \in \blacksquare^{d \times r}$, $A \in \blacksquare^{r \times k}$ with rank $r = 8$ and scaling factor $\alpha = 16$. This drastically reduces trainable parameters to only **147,456 (0.18%)**, enabling rapid, energy-efficient domain adaptation.

| Hyperparameter / Metric | Value | Technical Impact |
|---|---|---|
| Base Model Architecture | DistilGPT2 (Causal LM) | Lightweight, low-latency autoregressive transformer (82M params) |
| LoRA Rank (r) & Alpha (a) | r = 8, a = 16 (Scaling = 2.0) | Low intrinsic rank captures department FAQ structure with minimal drift |
| Target Adapter Modules | c_attn (Query/Key/Value Attention) | Focuses parameter updates on self-attention projection layers |
| Trainable Parameters | 147,456 / 82,060,032 (0.18%) | 99.82% parameters frozen, preventing catastrophic forgetting |
| Training Loss (Pre -> Post) | 9.8137 → 6.1742 | Substantial cross-entropy reduction across 4 training epochs |
| Perplexity (Pre -> Post) | 18,281.78 → 480.19 | Perplexity reduced by 97.4%, aligning syntax with academic inquiries |
| Training Hardware / Device | Apple Silicon MPS (Metal) | Fully hardware-accelerated local tensor computation in 7.2 seconds |

## Figure 1: LoRA Fine-Tuning Loss Convergence Curve



## 4. Unit U3 — Semantic Retrieval & Whisper Voice Input

**Dense Semantic Retrieval with Sentence-BERT:** Unlike fragile keyword search, the system encodes the department FAQ database using all-MiniLM-L6-v2 into 384-dimensional dense vectors. During query inference, the user prompt is projected into the same latent space to compute cosine similarities against both question and answer representations:

**Sim(q, d) = (u · v) / (||u|| ||v||)**

A hybrid lexical boost adds up to +0.20 for exact keyword overlaps (e.g., 'fees', 'attendance', 'hod', 'gpu'). Queries falling below the threshold (θ = 0.35) are flagged as out-of-domain or ungrounded.

**OpenAI Whisper Speech-to-Text Pipeline:** Spoken queries recorded via microphone or uploaded audio files are fed into OpenAI Whisper (base model). Audio is normalized, resampled to 16,000 Hz 16-bit mono PCM, and transcribed with zero external API dependencies. The resulting text seamlessly integrates into the multi-turn conversational memory.

## 5. Unit U4 — TTS Voice Synthesis & Event Poster Studio

**Text-to-Speech (TTS) Voice Engine:** Responses are converted to natural spoken audio via Google Text-to-Speech (gTTS) with an offline fallback to macOS native speech synthesizers. The TTS pre-processor cleans markdown asterisks, URLs, and code brackets to ensure smooth, pleasant auditory delivery.

**Automated Publication Poster Studio:** When users inquire about departmental events (e.g., *AI-Conflux Hackathon* or *NeurOlympics*) or request promotional flyers, the high-resolution Pillow graphic engine renders a 1080×1440 publication poster. It features cybernetic grid overlays, typography hierarchies, highlight badges, venue cards, and a procedural registration QR code.

### Figure 2: Publication-Ready Event Poster Generated for Department Hackathon




## 6. Unit U5 — Safety Guardrails, Hallucination Blocking & Zero-PII Policy

To ensure departmental compliance, strict three-tiered guardrails are enforced on every interaction:

- **Out-of-Scope Query Refusal:** Unrelated queries (politics, cooking recipes, sports, entertainment, personal counseling) are intercepted by regex classifiers and semantic distance checks, triggering an official academic refusal.
- **Hallucination & Unsupported Answer Blocking:** If an in-domain question lacks grounding documentation in the FAQ database (confidence < 0.35), the chatbot is strictly prevented from guessing or fabricating dates/numbers. It returns an official referral to Room 304.
- **Zero Personal Data Retention (Anti-PII):** User inputs containing phone numbers, emails, credit cards, or government IDs are scrubbed in real-time (replaced with [REDACTED_PHONE], [REDACTED_EMAIL]). No user data is ever written to disk.

## 7. Comprehensive Test Suite Validation Matrix

| Test Case / Unit | Input Query / Action | Expected Behavior | Validation Status |
|---|---|---|---|
| U1: System Prompt | Inspect get_system_prompt() | Contains persona, boundaries & tone | ■ PASS (Verified) |
| U2: Multi-Turn Memory | 2-turn sequential dialog | Preserves history across sliding buffer | ■ PASS (Verified) |
| U2: LoRA Convergence | 4-Epoch PEFT training | Loss drops 9.81 -> 6.17; PPL 18k -> 480 | ■ PASS (Verified) |
| U3: Semantic Retrieval | 'How much attendance needed?' | Cosine similarity > 0.50 for Doc #3 | ■ PASS (Score: 0.555) |
| U3: Whisper STT | 16kHz Speech Input | Transcribes clear voice prompt | ■ PASS (Verified) |
| U4: TTS Spoken Audio | Synthesize department greeting | Generates playable .mp3 audio file | ■ PASS (Verified) |
| U4: Poster Generator | AI-Conflux Hackathon details | Outputs 1080x1440 PNG graphic poster | ■ PASS (Verified) |
| U5: Out-of-Scope Refusal | 'How to bake a chocolate cake?' | Polite refusal & scope clarification | ■ PASS (Blocked) |
| U5: Anti-PII Redaction | Input: Phone & Email credentials | Replaces with [REDACTED_*] tokens | ■ PASS (Zero PII Saved) |
| U5: Hallucination Blocker | Ungrounded / made-up query | Refuses fabrication; directs to HOD | ■ PASS (Blocked) |

## 8. Live Application Universal URL & Deployment Guide

The project is structured for immediate worldwide access from any browser without complex firewall configurations:

- **Live Universal Application URL (Hugging Face Spaces Cloud):**
  **https://huggingface.co/spaces/gladiatorspring/FAQChatbot**

Provides 24/7 high-availability cloud hosting with public HTTPS accessibility from any desktop, tablet, or smartphone worldwide.

- **Official GitHub Repository:**
  **https://github.com/jeetdoesthings/FAQChatbot**

Contains complete source code, LoRA adapter weights, dataset JSON, test scripts, and demo instructions.

## 9. Conclusion

Project P24 successfully demonstrates a production-caliber Department FAQ Voice Chatbot satisfying all units U1 to U5. Through the fusion of dense Sentence-BERT retrieval, OpenAI Whisper speech recognition, LoRA parameter-efficient adaptation, speech synthesis, graphic design automation, and multi-layered safety guardrails, the chatbot delivers a responsive, reliable, and privacy-preserving virtual assistant for higher education departments.